Sparse Retrieval and BM25

1 Keyword retrieval.

BM25 is a ranking function that scores a passage by the frequency of the query terms in it and their rarity across the corpus. TF-IDF is a weighting scheme that multiplies how often a term occurs in a passage by how rare the term is in the corpus. An inverted index is a structure that maps each term to the passages that contain it.

2 Behaviour.

BM25 matches exact terms, so it handles acronyms and rare names well. Sparse retrieval fails when the query and the passage use different words for the same idea. BM25 needs no training and no vector index. Dense retrieval is a technique that maps text to vectors and ranks passages by their similarity to the query.

3 Term weighting.

TF-IDF rewards a term that is frequent in a passage and rare in the corpus. BM25 improves on TF-IDF by saturating the effect of a term that repeats many times. An inverted index lets BM25 find every passage that holds a query term without scanning the corpus. BM25 is the usual first stage of a hybrid retrieval pipeline together with dense retrieval.

4 Failure and fusion.

Sparse retrieval misses a passage that explains the idea in other words. Reciprocal rank fusion merges the BM25 ranking with a dense retrieval ranking without tuning weights. A cross-encoder can rerank the merged list because it reads the query and each passage together.